Why use two Load Balancers?

Because your Presentation and Application tiers have different responsibilities.

Presentation Tier

Handles:

- Browser requests
- HTML
- CSS
- JavaScript
- Images
- Frontend application

Architecture:

Internet
↓
Public ALB
↓
Frontend Target Group
↓
Frontend EC2 ASG

Application Tier

Handles:

- Business logic
- APIs
- Authentication
- Orders
- Payments logic
- Database communication

Architecture:

Frontend
↓
Internal ALB
↓
Backend Target Group
↓
Backend EC2 ASG
↓

RDS

This provides a strong security boundary: the backend servers don't need to be directly exposed to the Internet.

Final architecture scope

External:

- Cloudflare DNS
- Cloudflare Proxy/CDN/DDoS protection
- Git
- GitHub

AWS:

- VPC × 2
- VPC Peering
- EC2
- EBS
- Elastic IP
- Internet Gateway
- Route Tables
- Subnets
- Security Groups
- Network ACLs
- Auto Scaling
- CloudFront
- S3
- RDS MySQL
- CloudWatch
- CloudWatch Logs
- CloudWatch Alarms
- SNS
- CloudTrail
- IAM
- Systems Manager
- Session Manager
- Parameter Store
- AWS Budgets
- Cost Explorer
- AWS Health Dashboard
- Trusted Advisor
- Resource Groups/Tags
- EBS Snapshots
- RDS automated backup
- AMI